

Cumulative I_{XUV} Relative to Earth

Less likely to have atmosphere

LTT 3780 b

GJ 3473 b

TRAPPIST-1 b

GJ 3929 b

TRAPPIST-1 c

TOI-1468 b

TOI-771 b

LHS 1478 b

LHS 1478 c

TOI-406 c

HD 260655 c

LHS 1140 c

TOI-244 b

TOI-198 b

LTT 1445 A b

Cosmic shoreline $I \propto v_{\text{esc}}^4$

Dark bare rock

Inconclusive

Future target

10^3

10^2

10^1

10

15

20

25

Escape velocity (km/s)

LHS 1140 b

More likely to have atmosphere

